(Please write your Enrollment Number) Enrollment No. 032

MID-TERMEXAMINATION
(CBCS)
Course Name: B.Tech, Semester:3rd Sem
(Oct, 2022)

Subject Code: BIT 201 **Subject: Database Management Systems**

Time : 1 ½ Hours **Maximum Marks : 30**

Note: Q. 1 is compulsory. Attempt any two questions from the rest.

Q1 **(2.5*4)**

(a) Why is the use of DBMS recommended?

(b) What is the difference between DROP and DELETE commands?

(c) Are NULL values in a database the same as that of blank space?

(d) What is the difference between a database and a database system?

Q2 **(5,5)**

(a)Create an ER Diagram for the following scenario:
Students (rollno, sname), Courses (courseno, cname), Faculty (id, fname). Assume details wherever necessary and state them.

(b) Using the above-designed ER Diagram, create a database and tables using appropriate constraints. Insert five rows of data in the tables.

Q3 **(5,5)**

(a) Convert the Cross Product Operation of Relational Algebra into Join operation by adding/using other operations.

(b) What is the difference between Projection operation and Selection operation. Why Project Operation is a redundancy elimination operation.

Q4 **(5,5)**

(a) What are the different types of relationships in ER Model

(b)Construct an example of a ternary relationship relationship and map it into relations.